

Arvind Pujari

EDUCATION

OCT 2020 - DEC 2024	MRes+PhD in Nanoscience and Nanotechnology (NanoDTC) <i>University of Cambridge, Cambridge, United Kingdom</i> , MRes Grade: Distinction Received the Cambridge International Scholarship, awarded to the top 80 incoming PhD students at Cambridge
AUG 2016 - JUL 2020	B.Tech in Metallurgical and Materials Engineering <i>Indian Institute of Technology Madras, Chennai, India</i> , GPA: 8.90/10 Awarded the Dr Shankar Dayal Sharma President of India Gold Medal as the undergraduate student (out of a class of 900) with the best all round performance in curricular and extra-curricular activities between 2016-2020

PROFESSIONAL EXPERIENCE

AUG 2026 -	Tenure-Track Assistant Professor, Institute for Materials Research, Tohoku University, Sendai, Japan - Leading an independent research group focusing on developing scalable operando techniques - Designated as an "International Excellence Researcher" as part of Tohoku University's \$2.5 billion University of Research Excellence (UREX) Program. Tohoku University is currently ranked 3rd in Japan and 102nd in the world according to the QS Rankings 2026. - Supported by one postdoctoral researcher and several co-supervised students - Leading collaborative research with groups across the world: Prof. Clare Grey (University of Cambridge), Prof. Stefan Oswald (TU Munich), Prof. Mahdi Hamidi (University of Alberta) - Guest editor for the journal <i>Micromachines</i>, for a special issue on "Advanced Electrochemical Devices and Systems"
FEB 2025 - AUG 2026	Electrochemical R&D Scientist, Illumion Ltd., Cambridge, UK - Developing use cases and advancing cell design R&D for Illumion's revolutionary charge photometry technology - Led workflow development with industrial partners (Zeiss, ThermoFisher) for correlative x-ray/e-beam imaging with optical microscopy leading to the publication of multiple application notes - Improved <i>operando</i> cell configurations for industrial customers, published as a first/corresponding author article - Developed a method to analyse particle level expansion in silicon in collaboration with the German chemical manufacturer Wacker GmbH., leading to a first/corresponding author publication. - Led R&D and customer surveys for a new patented industry-focused product to analyse electrode heterogeneity. - Designed and developed a variable temperature cell holder in the first 6 months, with multiple sales secured

RESEARCH EXPERIENCE

OCT 2021 - DEC 2024	PhD Project: Light-battery interactions for enhanced energy storage and operando spectroscopy PIs: <i>Prof. Neil Greenham (Physics), Prof. Michael De Volder (Engineering), University of Cambridge, UK</i> - Finished PhD in 3.25 years with 6 first-author publications (4 with I.F > 20) and 4 co-authored papers - Performed rigorous control experiments and introduced standardized testing protocols for photobatteries, a new class of devices which seek to store and harvest energy within the same architecture Pivoted project after 2 years to develop a new, inexpensive optical technique to probe mechanisms in batteries - Experienced with materials characterization, battery/solar cell fabrication, and electrochemical testing Supervised 3 Master's projects, 1 PhD project, & taught courses on engineering maths and Raman spectroscopy Oral presenter at the International Conference on Materials Chemistry (MC16) (2023) and MRS Fall (2024) - Invited peer reviewer for Advanced Energy Materials, ChemComm, ACS Energy Letters, and Advanced Functional Materials (Reviews on ORCID) - Organised an exhibit on the basics of electrochemistry at the 2023 and 2024 Cambridge Science Festival - Responsible for training and maintenance of the shared Raman facility in the Physics department (> 100 users) - Ran inductions for new starters & delivered regular talks on lab safety and waste disposal to my research group
JUN 2024 - AUG 2024	Operando confocal microscopy of structured battery electrodes PI: <i>Prof. Hajime Arai, Institute of Science, Tokyo</i> - One of only 15 students from the UK to receive a JSPS Summer Fellowship in order to conduct a nine week project at the end of my PhD at the Tokyo Institute of Technology - Worked with Japanese industrial partners to study electrode heterogeneity in laser patterned battery cathodes - Work from the project led to a first/corresponding author publication. Some data was used in another manuscript. - Assisted and mentored a masters student working on the degradation of lithium ion batteries
NOV 2020 - FEB 2021	Microfluidic analysis of exosome-protein interactions at single-molecule level PIs: <i>Prof. Tuomas Knowles (Chemistry), Prof. Roisin Owens (Chemical Engineering), University of Cambridge, UK</i> - Completed a 9 week interdisciplinary lab rotation on single molecule detection of exosomes - Demonstrated that the (DigitalISA) method pioneered by the Knowles group can be used for highly-sensitive detection of exosomes with a low limit of detection (5 pM). Contributed to a publication in <i>Nature Communications</i>

TEACHING

- Supervisor for 2nd year engineering mathematics module. **Taught small groups** of undergraduates vector calculus, probability, and linear algebra for 4 years and marked mock examinations.
- Delivered lectures on solar cells/batteries and interviewed internship candidates as part of university's **widening participation** program
- Directly supervised two PhD students and three masters projects on topics ranging from electrochemically modulated photodetectors to thin-film all solid-state batteries.

ORAL CONFERENCE PRESENTATIONS

- *Can You Make Batteries for Operando Optical Measurements Without Sacrificing Performance?*, MRS Fall, Boston (2024)
- *Decoupling Heat and Light In Aqueous Photobatteries*, 16th International Conference on Materials Chemistry, Dublin (2023)

FUNDING ACQUIRED

EPSRC Core Equipment Award 2024

- Co-wrote research grant worth £250,000 with Prof. Michael De Volder to acquire an ion miller as a part of university shared equipment facilities, to be used by researchers in various departments

Cambridge Philosophical Society Research Studentship

- Awarded £2,500 to carry out work for three additional months 'beyond the usual standard of a PhD.

AWARDS

JSPS Summer Fellowship

- The JSPS Summer Fellowship offers the opportunity for pre/postdoctoral researchers to receive an orientation on Japanese language and culture and research systems as well as pursue research under the guidance of host researchers at Japanese universities and research institutes over a period of two months during the summer

Cambridge International Scholarship

- The Cambridge International Scholarship covers tuition for the top 80 incoming PhD students at the University of Cambridge

EPSRC PhD Fellowship

- The EPSRC Fellowship covers PhD stipends for a period of 4 years at UKRI rates

INAE Innovative Student Projects Award

- The INAE Innovative Student Projects Award recognizes the 10 best final-year undergraduate projects across India.

B. Krishnamurthy Award

- The B. Krishnamurthy Award is awarded to the student with the best final year undergraduate project in Metallurgical and Materials Engineering at IIT Madras

Dr Shankar Dayal Sharma Gold Medal

- The Dr Shankar Dayal Sharma President of India Gold Medal is awarded to the undergraduate student with the best all-round proficiency in curricular and extracurricular activities in each graduating batch.

Sri K M Ramamurthi Prize

- The KM Ramamurthi Award is awarded for the best all-round performance in cultural, co-curricular and organizational abilities during the first 6 semesters of the B.Tech Program in the departments of Chemical, Civil, Mechanical, Metallurgical and Ocean Engineering.

Purdue Undergraduate Research Fellowship

- One of only 15 international undergraduate students awarded a fully funded fellowship to conduct 9 weeks of research at Purdue University in the USA.